

Requirement Document: Stock Delivery + Volume + 30WMA Screener → Excel Generator

1) Objective

Build an application that:

1. reads a **text file of stock tickers**
2. fetches required market data
3. computes **Delivery %**, **Volume > 36-day MA**, and **30-week WMA weekstart cross**
4. generates a formatted **Excel report** showing **last 60 trading days**, color-coded delivery cells, per-stock summary counts, and daily band totals—matching the visual logic shown in your screenshot.

This is an end-of-day / batch-style reporting tool optimized for fast scanning in Excel.

2) Scope

In scope

- Input: tickers from text file
- Output: Excel workbook with:
 - last **60** trading days displayed
 - delivery % + volume signal inside each day cell
 - color coding by delivery % thresholds
 - 36-day MA volume comparison (“Yes/No”)
 - highlight if **price crosses above 30WMA on weekstart**
 - **FNO tickers** styled grey (ticker metadata area)
 - per-stock counts across **last 45 days** based on color categories
 - per-day totals of each delivery band (optional header block, but required if matching screenshot’s top counts)

Out of scope (v1)

- Brokerage integration, real-time streaming
 - Strategy backtesting
 - Automatic sector/segment classification (unless supplied via mapping file)
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3) Users & Workflow

User: trader/investor maintaining watchlists.

Workflow

1. User updates `tickers.txt`
2. Runs tool
3. Tool outputs `Delivery_Report_<YYYY-MM-DD>.xlsx`
4. User opens Excel and scans:
 - delivery strength across days (colors)
 - volume confirmation (Yes/No)
 - weekstart 30WMA cross signals
 - count columns for quick ranking

4) Inputs

4.1 Required: Ticker list file

- Filename: `tickers.txt`
- Format: one ticker per line
- Rules:
 - ignore blank lines
 - ignore comments starting with #
- Example:
 - AARTIDRUGS
 - AJANTPHARM
 - ALKEM
 - # ABBOTINDIA is optional
 - ABBOTINDIA

4.2 Required: FNO membership file (for grey styling)

- Filename: `fno_tickers.txt`
- Format: one ticker per line
- Same parsing rules as `tickers.txt`
- Purpose: mark tickers that are in Futures & Options (FNO)

4.3 Optional: Ticker metadata mapping

If you want left-side columns like Segment (Small/Mid/Large):

- Filename: `ticker_metadata.csv`
- Columns (optional):
 - ticker
 - segment (Smallcap/Midcap/Largecap)
 - sector (PHARMA/API etc.)
 - is_fno (Y/N) (alternative to `fno_tickers.txt`)

- If both `fno_tickers.txt` and `ticker_metadata.csv` exist, `ticker_metadata.csv` overrides.

4.4 Required: Configuration (file or UI)

Tool must support external config (no code edits required):

- exchange (e.g., NSE)
- lookback_days_display = **60**
- evaluation_days = **45**
- volume_ma_period = **36**
- weekly_wma_period = **30**
- weekstart_day = Monday (configurable)
- delivery_bins (default thresholds):
 - Purple: ≥ 80
 - Dark Green: ≥ 60 and < 80
 - Light Green: ≥ 50 and < 60
 - Blue: ≥ 40 and < 50
 - White: < 40 or missing
- display_order:
 - show days in descending order (latest first) OR ascending (configurable)

5) Data Requirements

5.1 Daily fields required (per ticker, per trading date)

- date
- close_price (for weekly aggregation / signals)
- traded_quantity (daily)
- delivery_quantity (daily)

5.2 Derived values required

- $\text{delivery_pct} = (\text{delivery_quantity} / \text{traded_quantity}) * 100$
- vol_ma36 = 36-day simple moving average of `traded_quantity`
- $\text{is_high_vol} = \text{traded_quantity} > \text{vol_ma36}$ (Yes/No)
- weekly close series (week-ending close)
- wma30_weekly computed on weekly close series
- $\text{weekstart_cross_above_30wma}$ (signal)

5.3 History fetch window

To reliably compute indicators:

- Must fetch at least: `lookback_days_display + volume_ma_period + buffer`
- Recommended buffer: +20 trading days (holidays, missing data)
- For weekly WMA30, tool must fetch enough weeks:
 - minimum 30 weeks + buffer (recommended 10 extra weeks)

6) Calculation Logic

6.1 Delivery Percentage (mandatory)

For each ticker/day:

- If `traded_quantity > 0`:
 - `delivery_pct = (delivery_quantity / traded_quantity) * 100`
- Else:
 - `delivery_pct = null` (blank cell)

Important: “Traded Quantity + Delivery Quantity should be used to compute delivery percentage.”

Interpretation used by tool (as explicitly requested): **delivery percentage is based on traded qty and delivery qty**, i.e. `delivery_qty / traded_qty`. (If you meant an alternative formula, it must be explicitly specified; default is the standard delivery% definition above.)

6.2 Delivery Color Coding (mandatory)

Delivery cell background color based on `delivery_pct`:

- Purple: `>= 80`
- Dark Green: `>= 60 and < 80`
- Light Green: `>= 50 and < 60`
- Blue: `>= 40 and < 50`
- White: `< 40 OR missing`

Thresholds must be configurable.

6.3 Volume condition (mandatory)

Compute 36-day MA of traded quantity:

- `vol_ma36 = SMA(traded_quantity, 36)`
Then:
- `is_high_vol = (traded_quantity > vol_ma36)`

If MA is unavailable due to insufficient history:

- `set is_high_vol = N/A`

6.4 Weekly 30WMA weekstart cross (mandatory)

Goal: “If stock price has crossed 30WMA on weekstart highlight in different column.”

Definitions

- Weekly close = close price of the last trading day of each week
- Weekly WMA30 = weighted moving average over last 30 weeks of weekly closes
- Weekstart day = first trading day of week (default Monday, configurable)

Cross-above signal

- A cross-above occurs when:
 - previous week close < previous week WMA30 AND
 - current week close >= current week WMA30

When to display

- The signal is attributed to the **weekstart trading day** of that week in the Excel via a dedicated column (not inside each date cell).

7) Excel Output Specification

7.1 File naming

- `Delivery_Report_<YYYY-MM-DD>.xlsx`

7.2 Sheet structure

Minimum v1: one main sheet (e.g., PHARMA_API or Report).

Optional v2+: multiple sheets by sector.

7.3 Layout (columns)

Left-side columns (recommended order):

1. SEGMENT (optional if metadata provided; else blank)
2. Above 30WMA (Weekstart Cross) (signal column)
3. STOCK (ticker)
4. 60 date columns (trading days)
5. Summary columns:
 - `Count_Purple_DarkGreen_LightGreen` (out of 45)

- `Count_Including_Blue` (out of 45)

7.4 FNO styling (mandatory)

If ticker is in FNO:

- Apply **grey fill** to ticker metadata cells:
 - `SEGMENT`, `Above 30WMA...`, `STOCK`, and summary columns
- Do **not** override the date-cell delivery colors.

7.5 Date column headers

- Must show the date (format configurable: `DD/MM/YYYY` or `DD-MM-YYYY`)
- Columns represent **last 60 trading days**.

7.6 Per-day cell contents (mandatory)

Each ticker × date cell must contain:

- `Delivery% | Volume>36MA`

Exact string format:

- `"<delivery_pct>% | <Yes/No>"`

Examples:

- `82.4% | Yes`
- `47.1% | No`
- If `delivery_pct` missing: blank (white)
- If `delivery_pct` present but MA not available: `82.4% | N/A`

Coloring

- Background fill based on `delivery_pct` bins only.

7.7 Commentary row/block (mandatory support)

Provide a top row (or merged block) labeled `Commentary-` above date columns.

- Default: leave commentary cells blank for manual edits
- Optional: support `commentary.json` keyed by date for auto-fill

7.8 Daily band totals (recommended; required if matching screenshot behavior)

For each date column, compute counts across all tickers:

- count of Purple (≥ 80)
- count of Dark Green (60–80)
- count of Light Green (50–60)
- count of Blue (40–50)
- count of White (< 40 or missing)

Display these counts in a compact header block aligned with each date column.

8) Summary Count Column Logic (mandatory)

Counts are computed across the **last 45 trading days** (not 60).

8.1 Count 1: Excluding Blue and White

Count_Purple_DarkGreen_LightGreen (out of 45) =

- number of days in the last 45 where `delivery_pct` ≥ 50
(i.e., bins: Purple + Dark Green + Light Green)

Exclude:

- Blue (40–50)
- White (< 40 or missing)

8.2 Count 2: Including Blue, excluding White

Count_Including_Blue (out of 45) =

- number of days in the last 45 where `delivery_pct` ≥ 40
(i.e., Purple + Dark Green + Light Green + Blue)

Exclude:

- White (< 40 or missing)

Important: Counts must be computed from numeric `delivery_pct`, not from Excel fill color.

9) Functional Requirements (FR)

FR-1 Read tickers from `tickers.txt`.

FR-2 Read FNO tickers from `fno_tickers.txt` (or metadata override).

FR-3 Fetch daily market data required to compute delivery_pct and volume MA.

FR-4 Compute:

- delivery_pct per day
- vol_ma36
- is_high_vol (Yes/No/N/A)
- weekly close series
- weekly WMA30
- weekstart cross-above signal

FR-5 Generate Excel with:

- 60 trading-day columns
- delivery cell values formatted as Delivery% | Yes/No
- delivery bin colors
- FNO grey styling on metadata columns
- weekstart cross highlight column
- count summary columns (45-day logic)
- optional daily band totals block
- commentary row/block

FR-6 Logging & error tolerance:

- If one ticker fails, continue processing others.
- Produce a run log with failures and missing dates.

10) Non-Functional Requirements (NFR)

- **Performance:** Support 300+ tickers with 60 days display efficiently.
- **Reliability:** Missing data yields blanks; tool never crashes.
- **Configurability:** Thresholds and periods configurable without code edits.
- **Determinism:** Same data snapshot + same config = same output.
- **Portability:** Must run on Windows (Excel-first workflow).

11) Edge Cases

- Non-trading days: only use trading dates.
- traded_quantity = 0 → delivery blank.
- delivery_quantity missing → delivery blank.
- insufficient history for MA36 → show N/A.
- insufficient weekly history for WMA30 → no cross signal (show blank/N/A).

12) Acceptance Criteria (Testable)

1. Excel includes 60 trading-day columns populated per ticker where data exists.
2. Each date cell contains text in format `Delivery% | Yes/No/N/A`.
3. Background colors match delivery bins exactly.
4. FNO tickers are grey-highlighted in ticker metadata cells.
5. Weekstart cross-above-30WMA is correctly flagged and highlighted in its dedicated column.
6. Count columns match numeric delivery thresholds over last 45 days.
7. Tool completes even if some tickers fail; log captures failures.

13) Build Plan (Vibe-Coding Friendly, but still disciplined)

- **V0:** Excel generation with 60 days + `delivery_pct` value + delivery color bins.
- **V1:** Add MA36 volume + append `| Yes/No` in each date cell.
- **V2:** Add summary counts (45-day) + FNO grey styling.
- **V3:** Add weekly WMA30 + weekstart cross column highlight.
- **V4:** Add daily band totals header block + commentary import.

Each version must ship a real workbook and a small verification dataset/test script.
